

MISTRAL AI · VIBE HACKATHON · PARIS · JULY 18, 2026



# Vibe Pulse

Give your agent eyes · live ML training & data  
visualization, right in the terminal.

## THE GAP

# Terminal-first ML devs still **leave the terminal** to see a curve

- ▶ Vibe **Work** charts your data in Canvas. Vibe **Code** (CLI): nothing.
- ▶ Jupyter or TensorBoard just to check convergence = context-switch tax.
- ▶ No terminal coding agent does this natively.  
**None.**



LIVE DEMO · REAL DATA

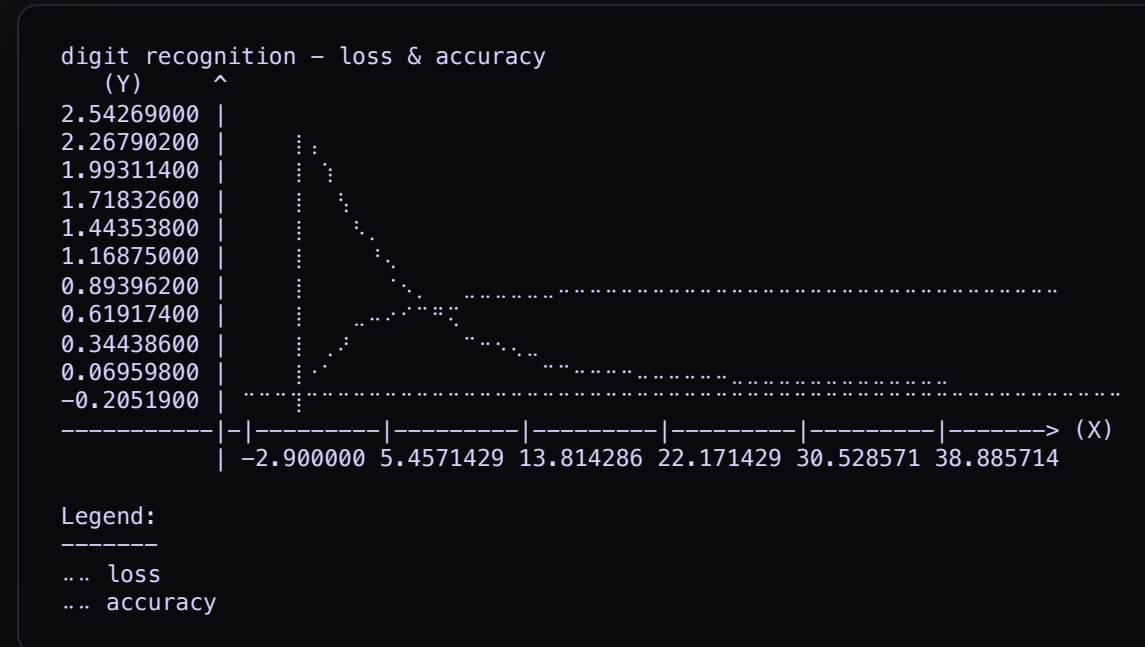
# Any data, straight in the terminal



- ▶ CSV, JSON streams, **Excel**, **SQLite** · one command each.
- ▶ The July 2026 heatwave, charted from a SQL query.
- ▶ Same charts, in color, **inside the Vibe conversation.**

IT ACTUALLY WORKS

# Machine learning, **fully** in the terminal

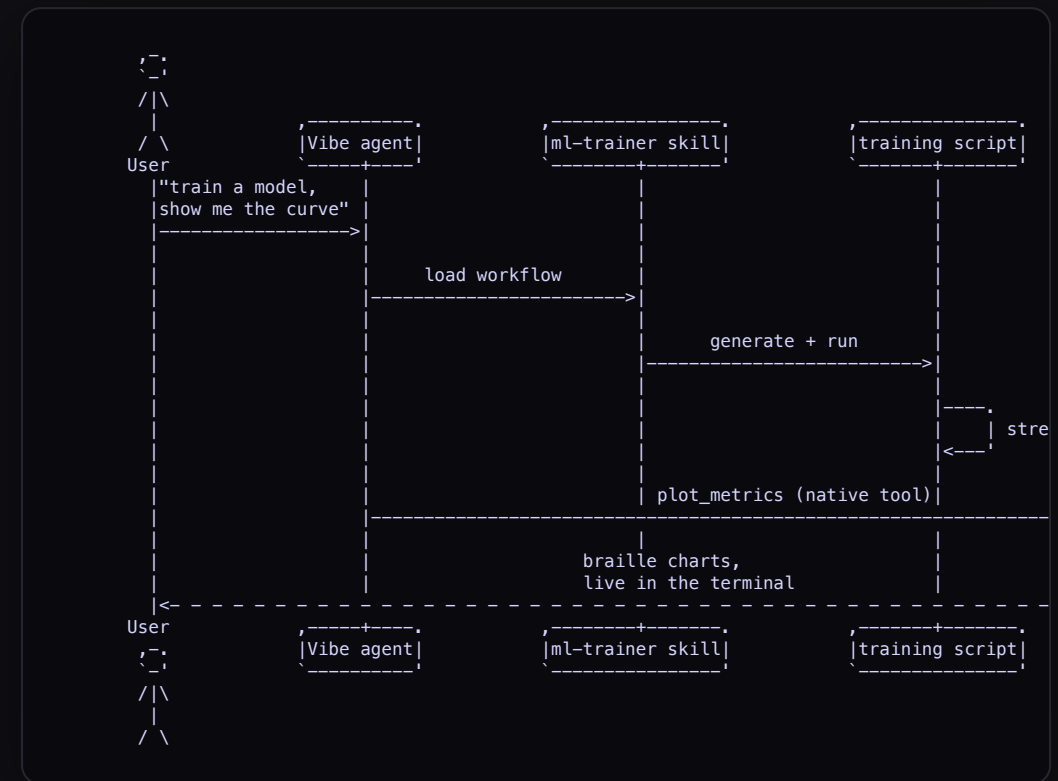


Digit recognition to **98 %**, curves drawn live epoch by epoch · confusion heatmap and the misclassified digits, as images, in the conversation.

## UNDER THE HOOD

# Every Vibe extension surface, **used**

- ▶ **Skill** · ml-trainer orchestrates the loop
- ▶ **Native tool** · plot\_metrics, 8 chart kinds, colored TUI widget
- ▶ **Provider** · Vibe inference on Qualcomm Cloud AI 100
- ▶ **Hooks** · post\_tool telemetry + pre\_tool secrets-guard



## WHO IT SERVES

# Three engineers, **one terminal**



### **Data scientists / ML engineers**

live training curves, confusion heatmaps, run tracking & comparison, agent-driven sweeps · no Jupyter, no TensorBoard.



### **Software engineers**

PlantUML architecture diagrams as ASCII art in the console (Mermaid next), agent telemetry · dev workflow, in place.



### **Embedded systems engineers**

compile + profile on real Snapdragon from the same session — 0.052 ms measured, not estimated.

# Same skill, both surfaces

same skill, both surfaces

Vibe Work · Canvas — weekly brief

## France Heatwave Report

Executive summary: severe heatwave, all major cities above 32°C, peaking at 39.6°C in Bordeaux.

### Highlights

- Peak: Bordeaux 39.6°C, Nantes 39.1°C, Lyon 37.6°C
- Coolest: Nice 32.0°C — still above summer norms
- All 11 monitored cities exceeded 32°C

### City Temperature Rankings

|    |          |        |
|----|----------|--------|
| 1. | Bordeaux | 39.6°C |
| 2. | Nantes   | 39.1°C |
| 3. | Lyon     | 37.6°C |
| 4. | Toulouse | 37.1°C |

### Highlights

- **Peak Temperature:** Bordeaux recorded the highest maximum temperature at 39.6°C, followed closely by Nantes at 39.1°C and Lyon at 37.6°C.
- **Lowest Reading:** Nice had the relatively coolest temperature at 32.0°C, though still well above normal summer averages.
- **National Impact:** All 11 monitored cities exceeded 32°C, confirming a widespread heatwave affecting the entire country.

### City Temperature Rankings

| Rank | City        | Temperature (°C) |
|------|-------------|------------------|
| 1    | Bordeaux    | 39.6             |
| 2    | Nantes      | 39.1             |
| 3    | Lyon        | 37.6             |
| 4    | Toulouse    | 37.1             |
| 5    | Montpellier | 37.0             |
| 6    | Paris       | 36.1             |
| 7    | Marseille   | 35.5             |
| 8    | Strasbourg  | 34.3             |
| 9    | Lille       | 32.8             |
| 10   | Nice        | 32.0             |

### Data Visualization

See [heatwave\\_temperatures.csv](#) for the underlying data and bar chart visualization.

### Recommended Action

Issue heatwave alerts for all regions, particularly Southwest France (Bordeaux, Toulouse) where temperatures are most extreme. Monitor vulnerable populations and ensure cooling centers are operational.

work-weekly-brief ran unchanged · canvas-to-cli rendered the terminal twin

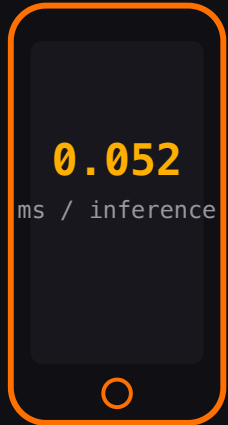
# W&B needs a server and a browser. **We don't.**

- ▶ **Run tracking** · every run logged to `runs/index.jsonl`
- ▶ **Run comparison** · `vibe-plot --compare` overlays any metric across experiments
- ▶ **Agent-driven sweeps** · “try 3 learning rates and compare” · no sweep server, the agent is the orchestrator
- ▶ **ASCII PlantUML diagrams** · architecture & dependency graphs without leaving the terminal
- ▶ **Canvas-to-CLI** · a Vibe Work skill converted unchanged (doc, table, chart twins)
- ▶ **vibe-arena** · LMArena-style model comparison in the console: Mistral vs OpenAI, latency & tok/s charted
- ▶ **Any data** · CSV, JSON streams, Excel, SQLite ( `--sql` ) · and Canvas-style tables ( `--table` )



REAL SILICON

# Trained in the terminal, **validated on Snapdragon**



**0.052 ms**

on-device inference  
Samsung Galaxy S24+ (AI Hub)

**24 MB**

peak memory on device

**Cloud AI 100**

Vibe's own inference runs on  
Qualcomm silicon (Cirrascle)

## TRACK 2 · GIVING BACK

# PR-ready upstream

- ▶ Draft PR **mistralai/mistral-vibe #920**: a native `plot_metrics` builtin
- ▶ Zero new dependencies · built on `render_braille` ...
- ▶ ...the primitive that animates **Vibe's mascot**. First product feature for the cat. 🐱
- ▶ Found & documented a real API pitfall (braille dot-index overflow) along the way

## ROADMAP

# The terminal as a **first-class** data science surface

- ▶ **Richer TUI** · textual-plotext RGB widgets, rich-pixels sample images, zoom/pan
- ▶ **Complex models** · CNNs, real datasets, fine-tuning, GPU runs · same live workflow
- ▶ **Fully local** · Ollama/llama.cpp provider is one config line away; a `/local` switch is in the works
- ▶ **Canvas-to-CLI** · full coverage (first Work skill converted today)
- ▶ **Deeper analysis** · dataset profiling, correlations, drift monitoring

VIBE PULSE

# Give your agent **eyes.**

Works today · a skill, a native tool, a provider, a hook.

[github.com/wilfred-dore/vibe-pulse](https://github.com/wilfred-dore/vibe-pulse) · PR [mistralai/mistral-vibe](#)  
#920